

Calculus Practice: Area and Volume Methods

Formula Reference Sheet

Area Between Two Curves

Use vertical slices for dx and horizontal slices for dy :

$$A = \int_a^b (\text{top } y - \text{bottom } y) dx, \quad A = \int_c^d (\text{right } x - \text{left } x) dy.$$

Find intersection points first. Split the integral whenever the top, bottom, right, or left boundary changes. Area is nonnegative, so reverse a subtraction or split at a crossing rather than keeping negative signed area.

Washers and Disks

A slice perpendicular to the axis of rotation produces a washer or disk:

$$V = \pi \int (R^2 - r^2) d(\text{variable}).$$

Here R is the distance from the axis to the farther boundary and r is the distance to the nearer boundary. For a disk, $r = 0$.

Axis of rotation	Washer slice	Usual variable
horizontal	vertical	dx
vertical	horizontal	dy

For rotation about $y = k$, radii are vertical distances such as $|f(x) - k|$. For rotation about $x = h$, radii are horizontal distances such as $|g(y) - h|$.

Cylindrical Shells

A slice parallel to the axis of rotation produces a shell:

$$V = 2\pi \int (\text{radius})(\text{height}) d(\text{variable}).$$

The radius is the distance from the slice to the axis. Shell height is top minus bottom for vertical shells and right minus left for horizontal shells.

Axis of rotation	Shell slice	Usual variable
vertical	vertical	dx
horizontal	horizontal	dy

Known Cross Sections

If every slice has a specified shape, first write its area A in terms of the base width, then integrate:

$$V = \int A(x) dx \quad \text{or} \quad V = \int A(y) dy.$$

Common cross-sectional areas are

Shape and given measurement	Area
Square, side s	s^2
Semicircle, diameter d	$\frac{\pi d^2}{8}$
Equilateral triangle, side s	$\frac{\sqrt{3}}{4}s^2$
Isosceles right triangle, leg ℓ	$\frac{1}{2}\ell^2$
Isosceles right triangle, hypotenuse h	$\frac{h^2}{4}$

The base width is found by top minus bottom for vertical slices or right minus left for horizontal slices.

Choosing a Method

1. Sketch the region and axis of rotation.
2. Decide whether vertical or horizontal slices avoid splitting or inverse functions.
3. Perpendicular to the axis means washers; parallel to the axis means shells.
4. Write every radius, height, or cross-sectional measurement as a distance.
5. Check bounds and units before evaluating.

Instructions

Complete in 60 minutes. Sketch each region, show the integral setup, and evaluate exactly. Choose dx or dy deliberately. For rotation problems, identify whether your slices produce washers/disks or shells before integrating.

Problems

1. Find the area enclosed by $y = 2x$ and $y = x^2$. Integrate with respect to x .
2. Find the area enclosed by $x = y^2$ and $x = 2y$. Integrate with respect to y .
3. Find the total area enclosed by $y = x^3$ and $y = x$. Your setup must account for the change in the upper curve.
4. The region bounded by $y = x^2$ and $y = 2x$ is revolved about the x -axis. Find the volume using washers.
5. The same region bounded by $y = x^2$ and $y = 2x$ is revolved about the y -axis. Find the volume using shells.
6. The region bounded by $x = y^2$ and $x = 2y$ is revolved about the y -axis. Find the volume using washers and integration with respect to y .
7. The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is revolved about the vertical line $x = 5$. Find the volume using shells.
8. In the first quadrant, the region bounded by $y = x^2$ and $x = y^2$ is revolved about the line $y = -1$. Find the volume using washers.
9. The base of a solid is the region bounded by $y = x$ and $y = x^2$. Cross sections perpendicular to the x -axis are squares. Find the volume.
10. The base of a solid is the region bounded by $y = x$ and $y = x^2$. Cross sections perpendicular to the x -axis are semicircles whose diameters lie in the base. Find the volume.
11. The base of a solid is the region bounded by $x = y^2$ and $x = 4$. Cross sections perpendicular to the y -axis are equilateral triangles. Find the volume.
12. The base of a solid is the region bounded by $y = 4 - x^2$ and $y = 0$. Cross sections perpendicular to the x -axis are isosceles right triangles whose legs lie in the base. Find the volume.